

WEEK-6

EXPLORATORY DATA ANALYSIS ON HEALTHCARE DATASET

Introduction

Exploratory Data Analysis (EDA) is used to understand and analyse a dataset using statistical methods and visualizations. In this analysis, a cleaned healthcare dataset is used to study patient information, medical conditions, billing amounts, insurance providers, admission types, and patient admissions.

Objective

The main objective of this analysis is to identify patterns and relationships in healthcare data. The analysis helps to understand how patient age, gender, medical conditions, insurance providers, admission types, and billing amounts are distributed.

Methodology

The dataset was loaded using **Pandas**, and different visualization techniques were performed using **Matplotlib** and **Seaborn**. The following analyses were carried out:

- Age distribution was analysed for different medical conditions using a **box plot**.
- Gender distribution for each medical condition was represented using a **bar chart**.
- Average billing amount for different medical conditions was calculated and visualized.
- Billing amounts were compared across different **insurance providers**.
- Billing amount distributions were studied using **violin plots**.
- Billing amounts were analysed based on different **admission types**.
- Monthly patient admissions were calculated and represented using a **line chart**.
- Correlation between **Age, Stay Days, and Billing Amount** was calculated using a correlation matrix and heatmap.

Results and Analysis

The visualizations provide a clear understanding of the healthcare dataset. The box plot shows the distribution of patient ages across different medical conditions. The gender bar chart helps compare the number of male and female patients for each condition. The billing analysis helps identify differences in average billing amounts across medical conditions and insurance providers. Violin and box plots provide information about the distribution and variation of billing amounts. Monthly admission analysis shows the variation in the number of patients admitted during different months. Finally, the correlation matrix helps understand the relationship between age, stay duration, and billing amount.

Conclusion

This EDA provides a better understanding of the healthcare dataset through statistical analysis and visualizations. It helps identify important patterns in patient demographics, medical conditions, billing amounts, insurance providers, and admissions. The analysis can be useful for further healthcare data analysis and decision-making.